



Final Project: Final Report

ICT 309 004 — Fundamentals of Artificial Intelligence

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Project Name: Professional Restaurant Reservation System

Student Names



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1. Idea

Our project is a **Professional Restaurant Reservation Chatbot System** designed to automate the process of booking tables in a restaurant using a conversational interface.

We chose this idea because traditional reservation systems often rely on manual processes, which can lead to errors such as double booking, incorrect data entry, and inefficient communication between customers and staff. By using a chatbot, we aim to simplify this process and provide a more user-friendly and efficient experience.

The system is implemented as a chatbot on Telegram, allowing users to interact with the reservation system through a familiar messaging interface. It collects important details such as the user's name, reservation date, time, and number of people, then checks availability before confirming the booking.

The chatbot also allows users to:

- View their reservations
- Edit booking details
- Cancel reservations

This makes the system more flexible and user-centered compared to basic booking systems.

2. How will the chosen project benefit you or society?

This project provides several benefits for both users and businesses.

For Customers:

- Easy and fast reservation process
- No need to call or visit the restaurant
- Ability to manage bookings anytime
- Clear confirmation and reduced confusion

For Restaurants:

- Reduces manual workload for staff
- Prevents double bookings
- Organizes reservation data efficiently
- Improves customer service

For Us (Learning Perspective):

- Helps us understand real-world system design
- Applies concepts from Artificial Intelligence such as decision-making logic and automation
- Improves problem-solving and teamwork skills

3. How did you do it?

We developed the system using a chatbot-based approach combined with database storage and logical validation.

System Flow (Chatbot Logic)

The chatbot follows a structured flow:

1. Ask for user name
2. Ask for reservation date
3. Ask for time
4. Ask for number of people
5. Check availability of tables and time slot
 - If available → continue
 - If not → ask user to choose another time/date
6. Ask user to confirm reservation
7. Save data into SQL database
8. Send confirmation message

Key Features Implemented

1. Smart Booking Flow

- Step-by-step interaction using structured logic
- Input validation at each stage
- Prevents invalid bookings

2. Availability Checking

- System checks if tables are available
- If full, booking is rejected and user must choose another option

3. Confirmation-Based Saving

- Data is only stored after user confirms
- Prevents incorrect entries

4. CRUD Operations

- Create → `/reserve`
- Read → `/myreservation`
- Update → `/edit <ID>`
- Delete → `/cancel <ID>`

Users can also search reservations using name or ID.

Database and Storage

- We used **SQL database** to store reservation data
- Data remains permanent even if the chatbot is turned off
- Ensures reliability and persistence

Challenges Encountered

One major challenge we faced is deployment:

- The chatbot currently **does not run 24/7**
- It only works when we manually run the code

To solve this, we plan to deploy the system using **Railway**

This will allow us to test whether the chatbot can:

- Run continuously
- Automatically respond to users

4. Who did what?

OUCH Phallyda

- Developed and implemented the **chatbot system**
- Designed the conversation flow and booking logic
- Integrated the chatbot with the database and handled core functionalities

SORN Vitou

- Contributed to **report writing, presentation slides, and system testing**
- Assisted in documenting system features
- Participated in testing the chatbot and identifying issues

KIN Kimcheng

- Contributed to **report writing, presentation slides, and system testing**
- Helped organize and design presentation content
- Participated in testing system functions and validating results

DORK Keovipheaviny

- Contributed to **report writing, presentation slides, and system testing**
- Assisted in structuring the report
- Participated in testing and improving overall system reliability

5. References (APA Style)

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